

ALVA BEACH, Australia

WMO: 952960

Lat: 19.4569S

Lon: 147.4833E

Elev: 8

StdP: 101.23

Time Zone: 10.00 (E10)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	12.4	2.1	4.4	17.9	4.9	5.4	16.5	9.9	21.2	9.3	21.5	3.2	210	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.2	31.9	26.0	31.2	25.5	30.7	25.2	27.7	29.9	27.1	29.5	26.6	29.1	5.5	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	22.9	28.9	26.5	22.0	28.6	25.8	21.1	28.3	88.6	29.8	85.7	29.5	83.3	29.2	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	10.3	9.3	8.0	34.1	1.1	1.9	7.2	35.5	6.6	36.6	6.0	37.6	5.2	39.0		
(5)			5.3	27.9	1.0	1.0	4.6	28.6	4.0	29.1	3.4	29.7	2.7	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.1	27.7	27.8	26.8	24.8	22.2	19.9	19.1	20.0	22.7	24.9	26.3	27.6
	DBStd	3.57	1.29	1.36	1.40	1.48	1.98	2.25	2.21	2.11	1.86	1.62	1.31	1.31
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	39	0	0	0	0	2	11	17	8	1	0	0	0
	CDD10.0	5154	549	497	520	443	378	297	283	309	382	461	490	546
	CDD18.3	2151	291	264	262	193	121	57	42	59	133	202	240	287
	CDH23.3	18730	3317	3031	2759	1560	632	145	62	142	539	1341	2098	3103
CDH26.7	4653	1092	1035	775	234	36	1	1	3	25	154	388	909	
Wind	WSAvg	5.2	5.6	5.3	5.6	5.3	4.5	4.2	4.3	4.7	5.3	5.8	6.2	6.0
Precipitation	PrecAvg	969	213	225	169	58	43	24	13	17	8	33	54	140
	PrecMax	1902	1049	709	548	400	219	197	62	100	63	250	228	600
	PrecMin	369	18	6	3	0	0	0	0	0	0	0	0	2
	PrecStd	395	204	162	148	85	51	38	15	26	14	54	62	146
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.6	33.5	32.4	30.4	28.7	26.7	25.6	27.0	28.5	29.7	31.2	32.2
		MCWB	26.6	26.4	25.7	24.4	22.9	21.6	19.8	19.5	21.7	23.5	24.7	26.2
	2%	DB	31.6	32.0	31.2	29.5	27.6	25.7	24.7	25.6	27.4	28.8	30.1	31.2
		MCWB	26.2	26.0	25.4	23.9	22.1	21.1	19.6	19.9	21.6	23.5	24.3	25.4
	5%	DB	30.9	31.2	30.5	28.7	26.8	24.9	24.1	24.8	26.7	28.1	29.4	30.6
		MCWB	25.6	25.6	25.0	23.5	21.6	20.5	19.2	19.6	21.2	22.9	23.8	25.0
	10%	DB	30.3	30.6	29.7	28.0	26.1	24.2	23.4	24.0	26.0	27.6	28.7	30.0
		MCWB	25.1	25.4	24.7	23.1	21.4	20.0	18.9	19.3	20.9	22.5	23.3	24.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.2	28.7	27.6	26.4	24.7	23.8	22.1	22.6	24.0	26.1	27.2	28.6
		MCDB	30.5	30.6	30.2	28.5	26.6	25.1	23.7	24.7	26.2	27.5	29.0	30.2
	2%	WB	27.4	27.7	26.8	25.6	23.7	22.6	21.3	21.7	23.2	25.1	25.8	27.3
		MCDB	30.0	29.8	29.4	27.6	26.1	24.3	23.2	23.9	25.6	27.1	27.9	29.4
	5%	WB	26.8	27.1	26.2	24.9	23.0	21.7	20.7	21.0	22.5	24.3	24.9	26.3
		MCDB	29.5	29.4	28.9	27.1	25.4	24.0	22.9	23.6	25.2	26.8	27.8	29.0
	10%	WB	26.2	26.5	25.6	24.2	22.3	20.9	20.2	20.3	21.9	23.6	24.2	25.6
		MCDB	29.1	29.0	28.5	26.8	25.1	23.5	22.6	23.0	24.7	26.4	27.5	28.7
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.2	6.0	6.7	7.9	8.2	8.6	8.4	6.5	5.4	4.7	4.6
		MCDBR	5.6	6.1	6.6	7.1	8.2	7.9	8.4	8.3	6.6	5.4	5.2	5.1
		MCWBR	2.3	2.6	2.8	3.2	4.1	4.3	4.7	4.3	3.0	2.4	2.1	2.1
	5% WB	MCDBR	5.1	4.9	5.7	6.0	6.7	6.4	6.5	6.7	5.6	5.0	4.8	4.8
		MCWBR	2.3	2.6	2.8	3.1	3.8	4.0	3.8	3.9	2.8	2.4	2.4	2.4
Clear-Sky Solar Irradiance	taub	0.404	0.398	0.390	0.367	0.349	0.337	0.325	0.336	0.362	0.390	0.394	0.403	
	taud	2.527	2.550	2.556	2.606	2.616	2.632	2.649	2.600	2.521	2.462	2.492	2.496	
	Ebn at Noon	942	936	920	904	885	878	902	922	932	932	945	943	
	Edh at Noon	113	109	104	92	85	81	81	91	106	117	116	116	
All-Sky Solar Radiation	RadAvg	6.44	5.97	5.54	4.99	4.28	3.87	4.33	5.15	6.14	6.73	6.91	6.89	
	RadStd	0.53	0.63	0.45	0.42	0.29	0.33	0.28	0.30	0.32	0.36	0.71	0.62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.13	-1.32	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page